UNIVERSITY OF UYO

DEPARTMENT OF MATHEMATICS

FIRST SEMESTER EXAMINATION

ADVANCED CALCULUS I

2023/2024 SESSION

MTH 313

TIME: 2 Hrs.

INSTRUCTION: Answer any four questions

1. $a.(i)$ Prove that $y = f(2x + at) + g(2x - at)$ satisfies the differential equation $4\frac{\partial^2 y}{\partial t^2} = a^2\frac{\partial^2 y}{\partial x^2}$

   (ii) If $x = r\cos\theta$, $y = r\sin\theta$, evaluate $\left(\frac{\partial r}{\partial x}\right)_y$ and $\left(\frac{\partial \theta}{\partial y}\right)_x$.

   $b.$ If $V = (x^2 + y^2 + z^2)^{\frac{m}{2}}$, find the value of $m(m \neq 0)$ which makes $\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$

2. $a.(i)$ Using the epsilon $(\in)$, delta (δ) approach, define the limit of a function of two variables $f(x, y)$ in terms of circular neighbourhood. 2

   (ii) Verify that $\lim_{(x,y)\to(0,0)} \frac{3x^2y}{x^2+y^2} = 0$ 2.5 4.5 each (11)

   $b.(i)$ state the conditions for a function of two variables $f(x, y)$ to be continuous at a point (x_0, y_0). 2

   (ii) Discuss the existence of the limit of the function $f(x, y) = \frac{xy}{\sqrt{x^2+y^2}}$ as $(x, y) \to (0,0)$. 4.5

3. $a.(i)$ Let u and v be the function of x and y. Prove that $\frac{\partial(u,v)}{\partial(x,y)} \times \frac{\partial(x,y)}{\partial(u,V)} = 1$

   (ii) If $x = uv$, $y = \frac{u+v}{u-v}$, find $\frac{\partial(u,v)}{\partial(x,y)}$

   $b.$ If $U = xy + yz + zx$, $V = x^2 + y^2 + z^2$ and $W = x + y + z$, determine whether there is a functional relationship between U, V, W and if so, find it.

4. $a.$ Define the Fourier series of periodic functions and determine the coefficients a_0, a_n and b_n 9

   $b.$ Find the Fourier series for the periodic function $f(x) = \begin{cases} 0, & -\pi < x < 0 \\ 1 & 0 < x < \pi \end{cases}$ 3 9 each (12)

5. $a.(i)$ Define the Laplace Transform of the function $f(t)$. (ii) Determine the Laplace transform of $f(t) = t^n$, where $t > 0$. Hence obtain the Laplace transform of t^3. 9

   $b.$ Find the Laplace transform of $f(t)$ defined as $f(t) = \begin{cases} \frac{t}{k}, & 0 < t < k \\ 1, & t > k \end{cases}$ 6 each (11)

6. Using Laplace transform , solve the initial value problem $y'' + 4y' - 5y = e^t$, $y(0) = 1$, $y'(0) = 0$

(13)

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